



BVR

Analysis 2 Quiz 1

26-08-2026

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1. $S = \mathbb{Z} \times (0, 1)$ with dictionary order: $(m, x) \leq (n, y)$ if $(m < n)$ OR $(m = n \text{ and } x \leq y)$.

(a). Exhibit a subset $T \subset \mathbb{R}$ (usual order) and an order preserving isomorphism $f: S \rightarrow T$. (b) Stating in two lines the reason, say whether S and $\mathbb{R}$ are order isomorphic. [4]

For (a) NO PROOFS Please; State your T and f .

a) Let $T = \mathbb{R} - \mathbb{Z}$

$$f: S \rightarrow T, \quad f(x, p) = x + p$$

(so it maps $f(3, 0.1)$ to 3.1)

b) S and $\mathbb{R}$ are ^{not} order isomorphic,
if $\mathbb{R}$ is not a disjoint union of countable open intervals

$$f(x, p) = \frac{x}{x+p}$$

or S is a disjoint union of countable open intervals, however $\mathbb{R}$ is not
 $\rightarrow S$ is disjoint union of countable open intervals I where $I \neq \mathbb{R}$.
 $\rightarrow \mathbb{R}$ cannot be written as such a disjoint union of open intervals I where $I \neq \mathbb{R}$.

practice writing solutions

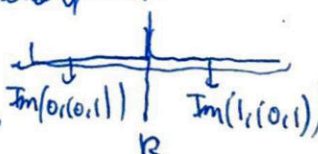
Suppose they were order isomorphic,

Let, consider the mapping of interval $(0, (0, 1)) \in S$. The image is disjoint from image of $(1, (0, 1))$, ord.

$$\text{Im}((0, (0, 1))) < \text{Im}((1, (0, 1)))$$

And there should be nothing in $\mathbb{R}$ between those two images due to isomorphism.

One of them must pick the boundary point, wlog, let

$\text{Im}((0, (0, 1)))$ pick the boundary point k . $\mathbb{R}$ is a continuum of $\text{Im}((0, (0, 1)))$. 

See its inverse image, say $(0, x) \in (0, (0, 1))$

$\rightarrow$ But $(0, x + \frac{(1-x)}{2}) > (0, x)$ so its image must be higher!

Contradiction! $\therefore S$ and $\mathbb{R}$ are not order isomorphic.

2. Consider real line R with usual distance. Let A be union of : interval $(0, 1)$; interval $(1, 2)$; rationals in the interval $(3, 4)$; the set $\{9, 10, 11, 12\}$.

$$A = (0, 1) \cup (1, 2) \cup [Q \cap (3, 4)] \cup \{9, 10, 11, 12\}$$

Then: (NO PROOFS Please, jut write your answer.)

[6]

(i) closure of A : $\bar{A} =$

$$\bar{A} = [0, 2] \cup [3, 4] \cup \{9, 10, 11, 12\}$$

(ii) interior of (closure of A): $(\bar{A})^o =$

$$(\bar{A})^o = (0, 2) \cup (3, 4)$$

(iii) closure of [interior of (closure of A)]: $\overline{(\bar{A})^o} =$

$$\overline{(\bar{A})^o} = [0, 2] \cup [3, 4]$$

(iv) interior of A : $A^o =$

$$A^o = (0, 1) \cup (1, 2)$$

(v) closure of (interior of A): $\overline{A^o} =$

$$\overline{A^o} = [0, 2]$$

(vi) interior of (closure of (interior of A)): $(\overline{A^o})^o =$

$$(\overline{A^o})^o = (0, 2)$$